FAKULTI TEKNOLOGI KEJURUTERAAN ELEKTRIK DAN ELEKTRONIK

BVI 2244 TECHNOLOGY OPERATION MANAGEMENT

REPORT - PROJECT

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TABLE OF CONTENTS

1 INTRODUCTION

1.1 Background of The Study

Operations management is a crucial discipline that focuses on the planning, organizing, and controlling of processes that transform resources into finished goods and services. In today’s competitive business environment, organizations must manage their operations efficiently to satisfy customer demand, minimize cost, and maintain high quality standards. This is especially important for retail businesses, where demand variability, inventory control, supplier reliability, and customer satisfaction directly affect overall performance.

In the pet retail industry, operational efficiency is highly dependent on accurate demand forecasting and effective inventory management. Products such as pet food and cat litter are classified as fast-moving consumer goods (FMCG), where frequent replenishment is required. Failure to accurately forecast demand can result in either stock shortages or excess inventory. Stock shortages lead to lost sales and dissatisfied customers, while excess inventory increases holding costs, storage requirements, and the risk of product damage.

This project focuses on applying key concepts of Technology Operation Management, including forecasting, capacity planning, project management, and quality management, to a real business scenario. By analyzing actual demand data from Kittencat Malaysia, students can understand how theoretical methods such as Moving Average and Linear Regression forecasting can be applied in real-world operations. These methods support better decision-making in inventory planning, supplier coordination, and capacity allocation.

Furthermore, the project integrates forecasting results with project management and quality improvement strategies. Rather than treating these components separately, the study demonstrates how demand forecasting can drive capacity planning decisions, which then justifies the implementation of improvement projects. Quality management tools such as the Fishbone (Ishikawa) Diagram are used to identify root causes of operational problems, ensuring that solutions are systematic and data driven.

Overall, this study aims to provide a comprehensive understanding of how effective operations management can enhance business sustainability, improve customer satisfaction, and support long-term growth, particularly for small and medium enterprises (SMEs) such as Kittencat Malayasia.

1.2 Company Overview

Kittencat Malaysia is a local pet retail business that operates in Pekan, a town located in the state of Pahang, Malaysia. The company specializes in providing a wide range of products and services related to cat care, making it a popular destination for cat owners in the area. As pet ownership continues to grow in Malaysia, the demand for reliable pet shops offering quality products has increased, positioning Kittencat Malaysia as an important service provider within the local community.

The core business activities of Kittencat Malaysia include the retail sale of cat food, cat litter, grooming accessories, toys, cages, hygiene products, and basic pet care items. Among these products, consumable items such as cat food and cat litter contribute significantly to the company's daily and monthly sales volume. These products require consistent inventory replenishment due to their high usage rate by customers.

Kittencat Malaysia is strategically located at **No.8 GF, Jalan Dagangan Peramu 1, Seri Dagangan Peramu, 26600 Pekan, Pahang, Malaysia**. The shop is easily accessible and situated in a commercial area, allowing customers from nearby residential and business zones to visit conveniently. The business operates seven days a week from **10:00 AM to 9:30 PM**, which increases customer accessibility and contributes to steady daily demand.

Customers can contact the business directly via telephone at **+60 11-3966 7338** to inquire about product availability, pricing, and promotions. This direct communication channel helps build strong customer relationships and improves service responsiveness.

As a growing business, Kittencat Malaysia faces several operational challenges common to small and medium enterprises, such as supplier dependency, limited storage capacity, manual inventory monitoring, and fluctuating customer demand. These challenges make effective forecasting and capacity planning essential to ensure smooth operations and prevent stock-related issues.

Despite these challenges, Kittencat Malaysia continues to expand its customer base due to its commitment to product quality, customer service, and competitive pricing. This makes the company a suitable case study for analyzing how structured operations management techniques can be applied to support business growth, improve efficiency, and enhance overall operational performance.

1.3 Objective of The Project

The objectives of this project are:

1. To analyze historical demand data for cat litter at Kittencat Malaysia.
2. To perform demand forecasting using Moving Average and Linear Regression methods.
3. To determine capacity requirements and identify potential bottlenecks.
4. To propose a relevant operational improvement project supported by WBS and Gantt Chart.
5. To identify a real quality issue and analyze its root causes using a Fishbone Diagram.
6. To propose a Quality Improvement Plan (QIP) to enhance operational performance.

1.4 Scope of The Project

The scope of this project covers:

- Demand forecasting for cat litter products.
- Capacity planning is based on forecasted demand.
- Project management planning to address operational issues.
- Quality management using the Ishikawa (Fishbone) Diagram.

2 FORECASTING AND CAPACITY PLANNING

2.1 Product Selection: Cat Litter

Cat litter is selected as the forecast product because it is a consumable item that requires frequent repurchases from customers. Demand for cat litter is relatively stable but influenced by factors such as promotions, supplier availability, and customer population.

2.2 Demand Data Collection

The demand data was collected for five consecutive months from Kittencat Malaysia.

Month	Demand (Unit)
July	718
August	669
September	554
October	700
November	790

2.3 Forecasting Methods I

2.3.1 Moving Average Method

The Moving Average method is an effective forecasting technique used primarily for stationary time series, where demand remains relatively stable without strong trends or seasonality. It works by calculating the average of a fixed number of the most recent actual demand values, which is then updated continuously as new data becomes available.

This method is useful for smoothing short-term fluctuations and providing a stable forecast. However, it has several limitations. The moving-average method is unable to accurately capture demand peaks and turning points, resulting in forecast lag.

When actual demand is persistently decreasing, the forecast tends to be overestimated, while during periods of continuous growth, the forecast is often underestimated.

Additionally, the method is not suitable for non stationary data that exhibit clear upward or downward trends. Since all historical observations are given **equal weight**, the method does not reflect the greater importance of more recent demand data. To overcome this limitation, a Weighted Moving Average method can be applied, where different weights are assigned to observations based on their recency.

2.3.2 Moving Average Calculation

Moving Average Formula

$$\text{Simple Moving Average} = \frac{(A_1 + A_2 + \ldots\ldots + A_n)}{n}$$

December Forecast base on the past five month period.

$$\text{December} = \frac{790+700+554+669+718}{5}$$

$$\text{December} = \frac{3431}{5}$$

December = 686.2 > 687 Unit

2.4 Forecasting Methods II

2.4.1 Linear Regression Method

Linear Regression is a forecasting method that identifies the relationship between time (independent variable) and demand (dependent variable). It fits a straight line through historical data points to represent the overall demand trend over a specific period.

The general equation for linear regression is:

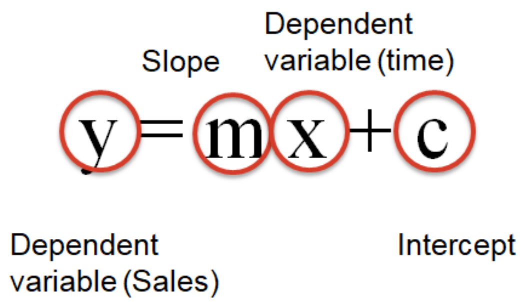
Y = mx + c

Where:

- **Y** = forecasted demand
- **m** = slope of the line, which represents the rate of change in demand
- **x** = time period
- **c** = intercept, which represents the demand level when x equals zero

This method assumes that the relationship between the variables is linear, meaning changes in demand occur at a constant rate over time. Linear regression is particularly useful when demand shows a consistent upward or downward trend, making it suitable for medium- to long-term forecasting and strategic planning.

2.4.2 Linear Regression Calculation



$$c = \bar{Y} - m\bar{X}$$

$$m = \frac{\Sigma XY - n\,\bar{X}\bar{Y}}{\Sigma x^2 - \; n\,\bar{X}^2}$$

December Forecast base on the past five month period.

Month (X)	Sale (Y)	XY	X̄	Ȳ	X^2
1	718	718			1
2	669	1338			2
3	554	1662	$\frac{1+2+3+4+5}{5}$ =3	$\frac{718+669+554+700+790}{5}$ =686.2	9
4	700	2800			16
5	790	3950			25
6					
		∑ XY =10,468			$\sum X^2$ =47

1. $m = \frac{10{,}468 - 5(3 \times 686.2)}{47 - 5(3^2)}$

   $m = \frac{10{,}468 - 10{,}293}{55 - 45}$

   $m = \frac{175}{10}$

   m = 17.5

Y6 = 17.5(6) + 633.7

Y6 = 738.7 > 739 Unit

2. $C = 686.2 - 17.5(3)$

$C = 686.2 - 52.5$

$C = 633.7$

2.5 Comparison of Forecasting Methods

Method	Characteristics	Suitability
Moving Average	Simple, smooth fluctuations	Suitable for stable demand
Linear Regression	Captures trend	Suitable for increasing demand

Linear regression is more suitable as it reflects the increasing demand trend for cat litter.

2.6 Capacity Planning

Based on forecasted demand, Kittencat Malaysia must ensure:

- Sufficient storage space
- Timely replenishment from suppliers
- Enough staff to handle stock movement

Potential Bottlenecks:

- Supplier delivery delays
- Limited storage capacity
- Manual stock monitoring

3 PROJECT MANAGEMENT

3.1 Project description and justification

KittenCat Malaysia currently operates as a retail pet shop that sell a pet related product. Market analysis shows that the demand for opening pet and grooming services is increasing and demanded because of customer busy with their own things, especially those who often outstation and worry about their pets. However, the current shop layout limits the company's ability to add new services which lead to capacity limitations and operational bottlenecks during busy periods.

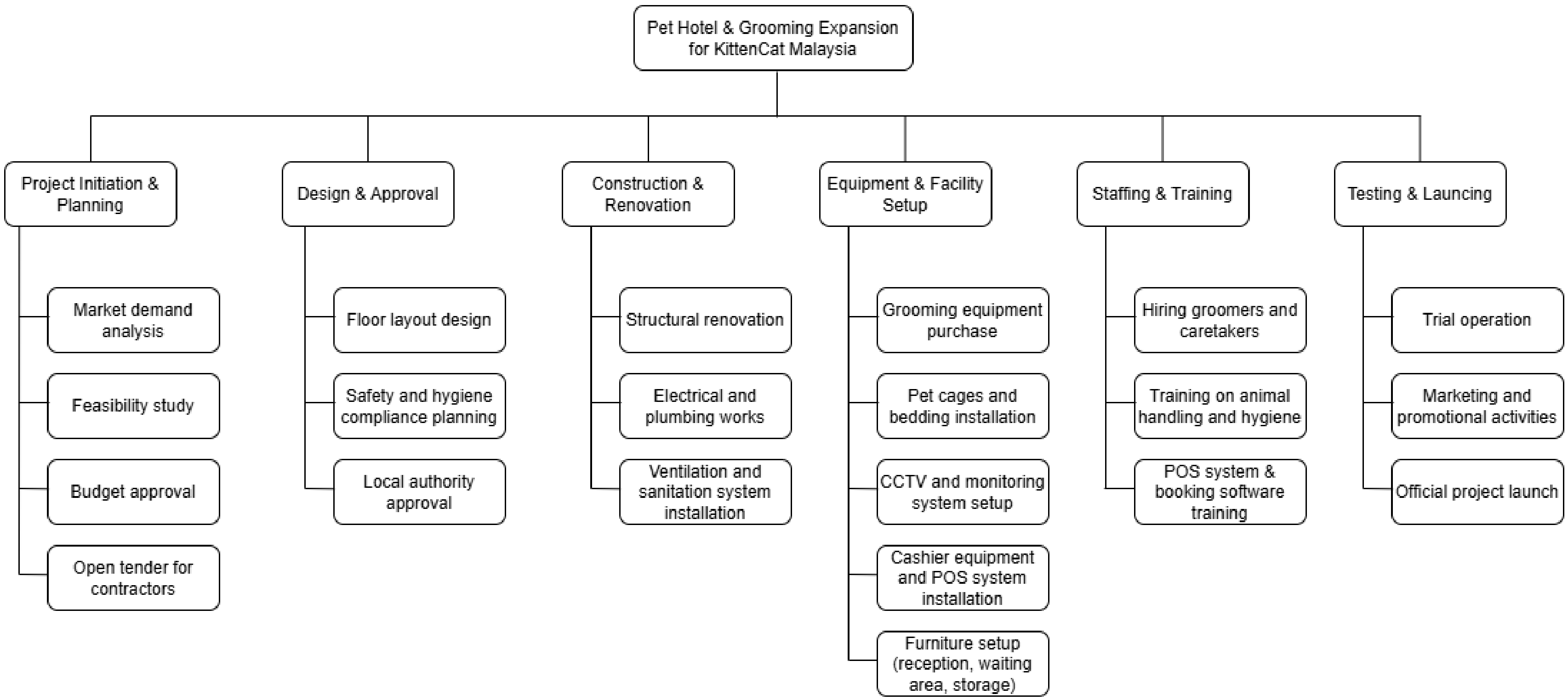
To solve this problem, KittenCat Malaysia plans to build a pet hotel and grooming facility above the existing retail shop. This expansion allows the company to operate pet hotel and grooming services while utilizing current premises without needing to relocate the business to another place.

This project is important because it help KittenCat Malaysia act to the growing demand for the pet hotel and grooming services. By adding these expansion services, the business can get more customers and ensure to reduce a long waiting time, especially during busy periods like weekends.

The expansion facility also can improve customer experience so that they can buy pet products, send their pets for grooming or hotel boarding services all in one place. Overall, KittenCat Malaysia will no longer depends on retail sales only and can get a new source of income by this expansion facility, also this project can improve their service quality, business capacity and customer satisfaction.

3.2 Work breakdown structure (WBS)

Here are the Work Breakdown Structure illustrate the KittenCat Malaysia for the Pet Hotel & Grooming expansion project, breaking down into specific tasks and subtask to ensure structured planning and efficient execution.



1) Project Initiation & Planning

This phase is the starting point of the project. Market demand analysis is carried out to understand customer needs and the demand for pet hotel and grooming services. A feasibility study is then conducted to check whether the project is practical in terms of cost, space, and operations. After that, budget approval is obtained to ensure sufficient funds are available. An open tender process is used to select suitable contractors in a fair and cost-effective manner before the project proceeds.

2) Design & Approval

In this phase, the project plans are prepared and approved. The floor layout design ensures that the space is arranged properly for grooming, pet boarding, and customer areas. Safety and hygiene compliance planning is done to meet animal welfare and health standards. Local authority approval is required to ensure the project follows building regulations and legal requirements before construction begins.

3) Construction & Renovation

This phase involves the physical work of the building. Structural renovation ensures the building is strong and safe for the new facilities. Electrical and plumbing works are carried out to support lighting, water supply, and grooming equipment. Ventilation and sanitation systems are installed to maintain cleanliness, control odors, and provide a comfortable environment for pets and staff.

4) Equipment & Facility Setup

This phase prepares the facility for daily operations. Grooming equipment is purchased for professional pet care services. Pet cages and bedding are installed to provide safe and comfortable spaces for boarding pets. CCTV and monitoring systems are set up to improve security and supervision. Cashier equipment and the POS system are installed to handle payments and bookings. Furniture such as reception counters, waiting area seats, and storage cabinets are also arranged to improve workflow and customer comfort.

5) Staffing & Training

This phase ensures that employees are ready to operate the new services. Groomers and caretakers are hired to handle pet care and grooming tasks. Staff are trained on animal handling and hygiene to ensure pet safety and cleanliness. Training on the POS system and booking software helps staff manage payments, reservations, and customer records efficiently.

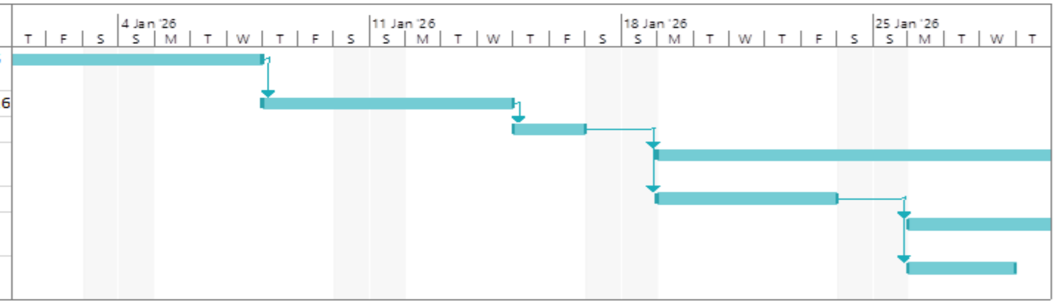
6) Testing & Launching

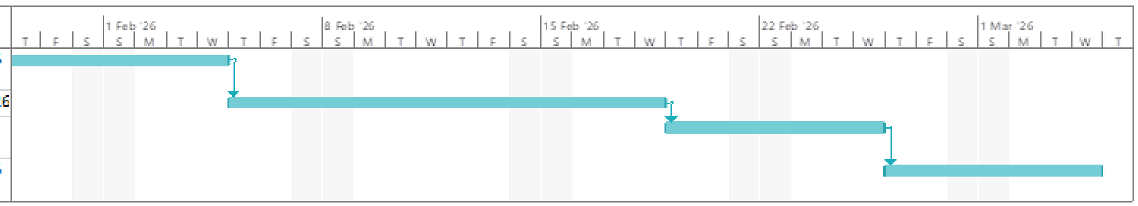
This final phase checks whether the facility is ready to operate. A trial operation is conducted to test equipment, staff performance, and service processes. Marketing and promotional activities are carried out to inform customers about the new services. The official project launch marks the start of full operations for the pet hotel and grooming facility.

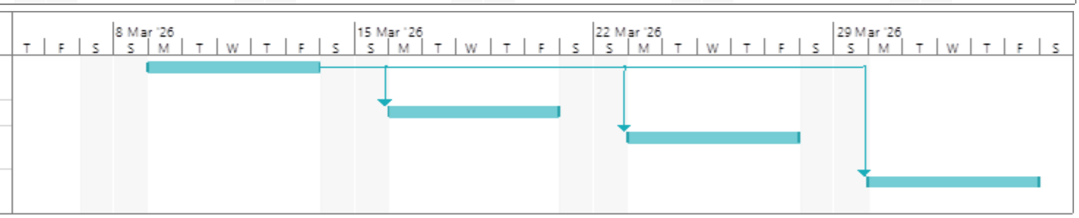
3.3 Gantt Chart

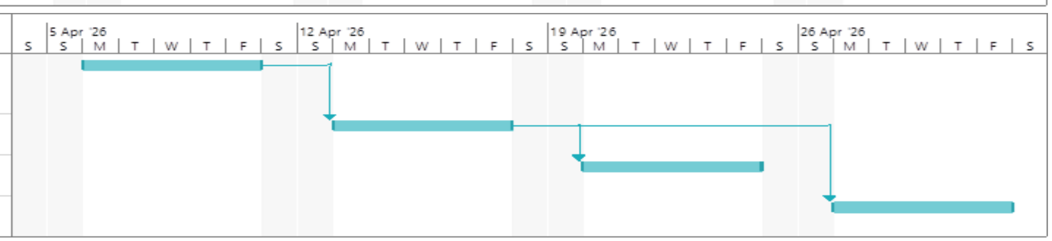
The Gantt Chart below show the detailed project schedule for the KittenCat Malaysia expand their business by adding Pet Hotel & Grooming Services. It shows all key tasks, their durations, start and finish dates, predecessors, and providing a clear presentation of the project timeline from 1st January 2026 to 25th May 2026.

ID	Task Mode	Task Name	Duration	Start	Finish
1		Market demand analysis	5 days	Thu 1/1/26	Wed 7/1/26
2		Feasibility study	5 days	Thu 8/1/26	Wed 14/1/26
3		Budget approval	2 days	Thu 15/1/26	Fri 16/1/26
4		Open tender for contractor	10 days	Mon 19/1/26	Fri 30/1/26
5		Floor layout design	5 days	Mon 19/1/26	Fri 23/1/26
6		Safety & hygiene complience	5 days	Mon 26/1/26	Fri 30/1/26
7		Local authority approval	3 days	Mon 26/1/26	Wed 28/1/26
8		Renovation material purchase	5 days	Thu 29/1/26	Wed 4/2/26
9		Structural renovation	10 days	Thu 5/2/26	Wed 18/2/26
10		Electrical & Plumbing works	5 days	Thu 19/2/26	Wed 25/2/26
11		Ventilation & Sanitation setup	5 days	Thu 26/2/26	Wed 4/3/26
12		Facility & grooming equipment purchase	5 days	Mon 9/3/26	Fri 13/3/26
13		CCTV & Monitoring Se	5 days	Mon 16/3/26	Fri 20/3/26
14		Pet Cages & Bedding Installation	5 days	Mon 23/3/26	Fri 27/3/26
15		Cashier equipment & POS system setup	5 days	Mon 30/3/26	Fri 3/4/26
16		Furniture setup (reception, waiting area, storage)	5 days	Mon 6/4/26	Fri 10/4/26
17		Hiring groomers & caretakers	5 days	Mon 13/4/26	Fri 17/4/26
18		Training on animal handling & hygiene	5 days	Mon 20/4/26	Fri 24/4/26
19		POS system & software booking	5 days	Mon 27/4/26	Fri 1/5/26

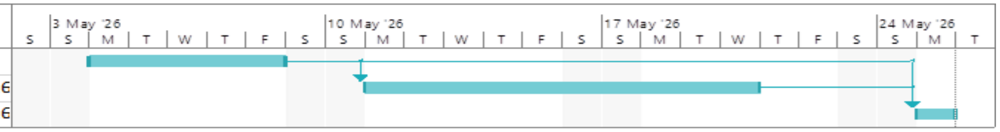








ID		Task Mode	Task Name	Duration	Start	Finish
20			Trial operation	5 days	Mon 4/5/26	Fri 8/5/26
21			Marketing & Promoti	8 days	Mon 11/5/26	Wed 20/5/26
22			Official Launch	1 day	Mon 25/5/26	Mon 25/5/26



3.4 Discussion on project impact on operational performance

Forecasting result for KittenCat Malaysia show that customer demand for cat litter sand product are stable with forecasted sales of 687, 680 and 663 units for the next three months. For now, KittenCat Malaysia faced an operational bottleneck because of the business only depends on retail sales of cat products that limiting their business growth potential. The proposed project to reduce this limitation is expanding the business to offer pet hotel and grooming services. This project helps them to reduce bottlenecks by creating a new source of income and increasing business capacity. It also can attract more customers to come to their shop to buy their pet product while they can experience the pet hotel and grooming services, also provides a more stable cash flow through the expanded business. In addition, the new services meet the growing customer demand for convenient and reliable pet care. Overall, the project improves operational performance and helps KittenCat Malaysia meet forecasted demand effectively.

4 MANAGING QUALITY

4.1 Identification of quality issue

The selected company for this study is Kittencat Malaysia in Pekan, Malaysia. A pet food retail shop that depends heavily on external suppliers for product availability. Based on the forecasting, capacity planning, and project development analysis, one major quality issue identified in the company is unstable food stock availability.

The food stock inconsistency occurs mainly due to supplier related constraints. The supplier was threatened to deliver only one single outlet or else facing losing client and breaching contract.

4.2 Root cause analysis (using any QC tools)

To analyze the root cause of that unstable food stock availability. We use a Cause-and-Effect Diagram (Fishbone Diagram or Ishikawa). This QC tool helps identify a potential cause by categorizing some major contributing factors.

Fishbone (Ishikawa) Diagram

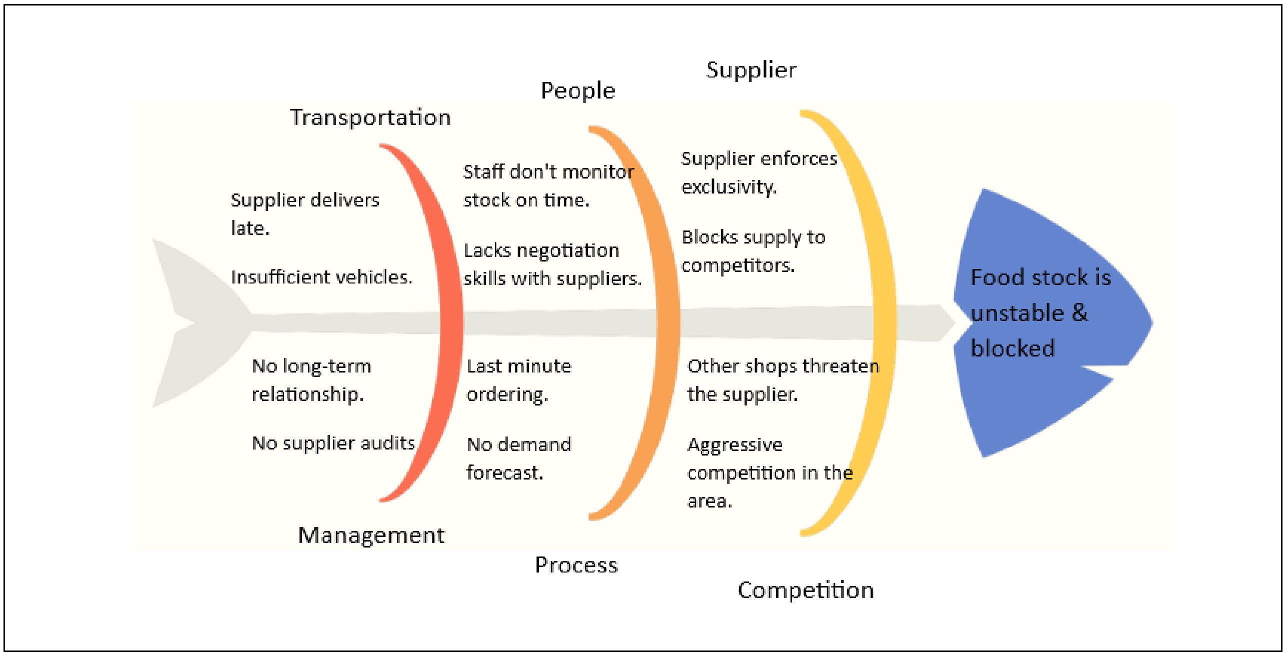


Figure 1: Fishbone (Ishikawa) Diagram of Kittencat

Effects:

- Food stock unstable and frequently blocked

Root caused categorized and analysis:

The cause-and-effect analysis identified multiple contributing factors including transportation delays, people related issues, supplier constraints, management weakness, and process inefficient. However, competition acts as a dominant root cause. Intense competition among shops and retailers has led suppliers to prioritize certain shops and enforce an exclusive supply arrangement otherwise suppliers will breach the contract and agreement if notices supplying multiple shops and retails.

Effect of this quality issue:

- Shop will run out of stock
- Customer will lose its interest (move to other shop)
- Costs increase (in term on getting food supply)
- Shop reputation or image will drop (not getting the used or frequent food product)

4.3 Quality Improvement plan (QIP)

To address the identified quality issue, a Quality Improvement Plan (QIP) is proposed to improve supplier reliability, inventory control, and demand planning.

Root cause category	Improvement Action	Responsible Party	Expected Outcome
Competition	Diversity supplier to avoid exclusivity dependence	Management	Reduced risk of food supply blockage
	Develop own product	Management	Lower dependency on supplier
Supplier	Establish long term supplier	Management	A stable and committed supply
	Get a backup supplier list	Management	Faster recovery during shortage
People	Train staff on stock monitoring and inventory control	Management	Improved stock counting
	Improve negotiating skills with supplier	Management	Better supplier cooperation
Transportation	Identify alternative logistics providers	Management	Reduced delivery time delay
	Plan delivery frequency based on demand trends	Management	Consistent stock arrival
Management	Develop supplier risk management strategy	Management	Proactive risk mitigation
	Review policies and do improvement	Management	Strong strategic decision
Process	Implement basic forecasting when to order	Management	Accurate ordering quantities
	Set minimum stock level	Management	Early restoration action

The Quality Improvement Plan supports the company's inventory planning and supplier management by ensuring consistent product availability. Special emphasis is placed on competition risks, as competitive pressure was identified as the primary factor affecting food stock availability. The proposed improvement actions aim to stabilize supply flow, strengthen internal inventory and procurement processes, and enhance Kittencat's overall operational resilience and competitiveness.

5 CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This project has successfully demonstrated the application of concepts in a real business environment by using Kittencat Pekan as a case study. By analyzing historical demand data for 5 consecutive months as cat litter is our main focused product. The study highlighted the importance of accurate demand forecasting to support effective inventory and planning order decisions.

We are using two forecasting methods which are moving linear methods and linear regression methods to forecast future demand. The comparison between the two of them showed that the linear regression method is a more suitable method as it captures the trend of demand more effectively. The forecasting methods result in a strong foundation for inventory planning, supplier coordination and storage management, reducing the risks of stock shortage and excess inventory.

A project management was developed for the expansion of pet hotel and grooming services. The project was justified based on market demand, operational limitations, and business growth opportunities supported by a work-based structure (WBS) and Gantt chart. A major quality issue related is unstable food stock availability as we are using the Fishbone (Ishikawa) diagram to identify and analyze the issue. The root cause analysis highlighted competition and supplier exclusivity as the dominant factors that affect stock availability.

In capacity planning analysis revealed that several operational bottlenecks were identified including supplier issues, limited storage capacity, and manual stock monitoring. To address these issues, a quality improvement plan (QIP) was developed to improve supplier reliability, inventory control, and demand planning.

Overall, this project illustrates how the integration between forecasting, capacity planning, project management and quality management that can enhance operational efficiency, improve customer satisfaction and strengthen business sustainability in the future.

5.2 Recommendations

Based on the findings of this study, several recommendations are proposed to further improve Kittencat Malaysia's9 operational performance:

1. **Using advanced forecasting techniques**
   Currently, we are using moving average and linear regression methods to forecast our demand. We are suggesting using weighted moving average as this method will improve forecast more accurately by giving more emphasis to recent demand data.
2. **Strengthen supplier management strategy**
   We are suggesting that the company should reduce dependency on a single supplier by diversifying suppliers and establishing long-term agreements with multiple vendors. This approach will minimize supply disruption risks caused by competition and exclusivity arrangements.
3. **Implementing minimum stock level policies**
   By setting a safety stock and reorder point levels based on forecasted demand will help prevent stockouts and ensure timely replenishment, especially for fast-moving items such as cat food and cat litter.
4. **Continuous Quality Improvement**
   There are many types of quality management tools we can use to identify and address operational issues. So, the review of supplier performance, delivery reliability, and internal processes should be conducted.

By implementing these recommendations, Kittencat Malaysia can improve its operational resilience, maintain consistent product availability, enhance customer satisfaction, and support sustainable long-term business growth.

6 REFERENCES

What is a Fishbone Diagram? Ishikawa Cause & Effect Diagram | ASQ

Quality Improvement Plan Development